EXPERIMENTAL INVESTIGATION OF THE CHEMISTRY OF EXCITED STATES OF RARE GASES

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Second Quarterly Progress Report

The personnel working on the E(11-1)-2807 ERDA contract are Mr. J. E. Velazco (graduate student) and Dr. J. Rodriguez (postdoctoral fellow). Mr. Velazco has done the work described in the attached manuscript (rough draft form). Mr. Rodriguez has been developing new techniques for study of the chemistry of rare gases.

The measurement of the total quenching rate constants and the branching ratios for XeF or KrF formation from reactions of $Kr(^3P_2)$ and $Xe(^3P_2)$ metastable rare gas atoms with fluorine containing molecules is completed. A summary of the work is provided in the Abstract of the manuscript and need not be repeated. Mr. Velazco will do a similar (but less extensive) study of $Xe(^3P_2)$ with other halogen donors (RI, RBr and RCl) in the next few months. He also will be writing his Ph.D. Dissertation before Oct. 15, 1976.

Dr. Rodriguez has assembled a vacuum ultraviolet monochromator system which will be used for studying the reactions of $Ar(^3P_2)$ with RF compounds (the ArF emission covers the 210-170 nm region). Measurement of the branching ratios for ArF formation is essential to relate our prior studies of ArCl (J. Chem. Phys. In Press (1976)) with the KrF and XeF work described in the attached manuscript. We also intend to use the vacuum U. V. monochromator for preliminary studies of rare gas excimer chemistry.

Dr. Rodriguez also has improved a tunable dye laser apparatus. The dye laser is used to pump argon atoms from the metastable $(4s,^3P_0)$ or $(4s,^3P_2)$ states up to the Ar*(4p) or Ar*(5p) levels. The subsequent fluorescence from these levels is used to deduce total decay rate constants for these Ar*(4p) and Ar*(5p) levels in an argon bath gas. Recent work has resulted in (i) reduction of the scattered laser light from the windows of the fluorescence cell and (ii) improvement of the metastable source so that higher pressure ($\sim$20 torr) can be obtained. The two body decay rate constants have been measured and we will be able to, at least, establish upper limits to the three body rate constants for selected levels of the Ar(4p) and Ar(5p) manifolds.

WARNING

This manuscript is still in draft form and the tabulated results may change by 20% depending upon finalization of calibration procedures. With benefit of good fortune the manuscript will be finalized by May 15 and will be submitted to J. Chem. Physics for review.

D. W. Setser

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QUENCHING RATE CONSTANTS FOR METASTABLE ARGON, KRYPTON AND XENON ATOMS BY FLUORINE CONTAINING MOLECULES AND BRANCHING RATIOS FOR XeF AND KrF FORMATION

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Abstract

The flowing afterglow technique was used to study the reactions of $Xe(^3P_2)$, $Kr(^3P_2)$ and $Ar(^3P_2)$ metastable atoms with small fluorine containing molecules. Fluorides from Groups III through VIII (XeF_2) were examined. Although all the fluorides have large quenching rate constants, only F_2, XeF_2, some interhalogen fluorides and small molecules with the OF or NF bond have high branching ratios for XeF or KrF excimer formation. The branching ratio measurements were made via comparison to the XeCl and KrCl emission intensities from reaction of $Xe(^3P_2)$ and $Kr(^3P_2)$ with Cl_2, which were adopted as reference reactions. Within experimental error, the branching ratios are unity for $Kr(^3P_2)$ and $Xe(^3P_2)$ with Cl_2, F_2 and OF_2. Variation of argon pressure in the 1 - 40 torr range shows extensive vibrational relaxation but no electronic quenching of the lowest energy excimer state of XeF*, XeCl*, KrCl* or KrF*. The occurrence of vibrational relaxation suggests a spontaneous radiative lifetime of > 50 nsec. A higher energy excimer state ($^2\Pi_{1/2}$) was identified for XeF, XeCl, XeBr, XeI, KrF, and KrCl. Discussion is presented regarding the chemical dynamics and chemiluminescence of these reaction systems.

1. INTRODUCTION

Since our initial report of the xenon-halide diatomic excimer emissions[1] from the two body reaction between metastable $Xe(^3P_2)$ atoms and halogen containing molecules, interest in these and similar systems has grown extremely rapidly. In independent work, Golde and Thrush[2] reported the vacuum ultraviolet emissions associated with ArCl, ArBr and ArO. Subsequently this[3,4] and other laboratories[5-7] have identified all of the other Ar, Kr and Xe - halogen excimer emissions, except for Ar-I* which may be totally predissociated. Golde[5] has given a general discussion of these bound-free excimer emissions and tabulated some spectroscopic parameters based upon his interpretations. Conventional spectroscopic measurements from high pressure XeCl spectra have established parameters for both the upper and lower states of XeCl.[8] In addition to the intrinsic interest in these new examples of excited state chemistry, the bound-free nature of the transitions provides excellent possibilities for laser activity. Indeed, using e-beam pumping of rare gas-halogen mixtures at total pressures ranging from 0.5 to several atmospheres, laser emission has been reported for XeF (from Xe + F_2, Xe + NF_3),[10,11] XeCl (Xe + Cl_2),[8,10,11] XeBr (Xe + Br_2)[9] and KrF(Kr + NF_3, Kr + F_2),[10,11] with estimated efficiencies of $\leq$ 15%.[11] Laser activity also has been observed for these systems using electric discharge excitation.[12]

An understanding of the mechanisms responsible for emissions in the laser systems requires the knowledge of rate constants and branching ratios for the relevant steps; therefore, our laboratory has undertaken the task of systematically measuring the total quenching rate constant and the branching ratios for reaction of $Ar(^3P_2)$, $Kr(^3P_2)$ and $Xe(^3P_2)$ with a variety of halogen donors. In order to meet criteria for a satisfactory laser, certain properties of the halogen donor, RX, are desirable: 1) The reaction M* + RX → MX* + R must be efficient, i.e. the branching ratio for formation of MX must be large. ii) There should be little absorption by RX at the laser frequency. iii) RX

should be as non-toxic and non-corrosive as possible.

The metastable rare gas atoms react directly[1-4] with halogen containing molecules to form the excimers, although other processes also may contribute to the formation under laser excitation conditions. The mechanism for direct excimer formation involves electron transfer with formation of M^+RX^- followed by dissociation of R; two obvious criteria in the selection of the halogen donor are that RX should have a large electron affinity and that $D_o^\circ(RX) \lesssim$ 70 kcal $mole^{-1}$. All molecules readily quench the Xe, Kr and Ar metastable states;[13,14] the mechanisms for interaction are varied and efficient formation of MX can be expected only for a small group of halogen donors.[4] The present work shows only F_2, some simple interhalogens,and molecules with NF or OF bonds to be good fluorine donors. With interhalogen compounds, excimers of both halogens are frequently observed. These results parallel previous findings from $Ar(^3P_{0,2})$ + RCl systems.[4] In addition to identifying good donor molecules for laser applications, the present work represents further effort at providing a data base that will eventually facilitate the understanding of the interaction of electronically excited rare gas atoms with reagents of large electron affinity.

This paper reports rate constants and branching ratios from reactions of fluorine containing molecules with $Xe(^3P_2)$ and $Kr(^3P_2)$, and rate constants for the $Ar(^3P_2)$ reactions. Forthcoming papers will deal with other halogens. Analysis of the excimer emission spectra also permits some conclusions to be made regarding the upper states of XeF, XeCl, KrF and KrCl. The organization of the paper is as follows: Section 2A describes the experimental method for determining the total quenching rate constants, while section 2B describes the emission measurements used to determine the branching ratios. Section 3A summarizes the rate constants for the quenching of Ar, Kr & Xe metastable atoms by fluorine containing molecules; the results of the branching ratios are

summarized in Sections 3B and 3C. Finally in section 4 all these results are discussed. Of particular interest is the observation of vibrational relaxation at pressures above ∿ 2 torr which suggests that the lifetimes are longer than ∿ 50 ns. The spectroscopic results for XeF and XeCl are compared to those presented in a recent paper by Brau and Ewing[7] and by Tellinghusen and co-workers.[8] An emission previously observed[7] at ∿260 nm is confirmed to be another excited state of XeF. The corresponding state of XeCl, XeBr, XeI, KrCl and KrF also is identified.

2. EXPERIMENTAL

2A. Total Quenching Rate Constants

The flow reactor (pyrex glass tube with attached windows) and the flowing afterglow technique used in the determination of the total quenching rate constants are the same as previously employed.[13,14] Metastable Ar atoms are produced by passing a flow (0.152 moles/min) of Ar, purified by two low pressure molecular sieve traps maintained at 77°K, through a hollow cathode discharge (250 V, ∿1 mA) located approximately 15 cm upstream from the reagent inlet. Linear flow velocities of the order of 8000 cm/sec in a 3.1 cm I.D. tube are achieved by a combined Roots blower and a 1000 l/min vacuum pump. The experiment consists of monitoring (atomic absorption spectroscopy) the metastable atom concentration along the tubular reactor for different reagent concentrations. Light from a Pen-Ray lamp (Oriel Corp.) crosses the reactor perpendicular to the gas flow and, after being mechanically chopped at 600 Hz (PAR 125), enters a 0.3 m McPherson monochromator which selects the Ar, Kr or Xe line of interest (Ar, 811.5 nm; Kr, 811.2 nm; Xe, 881.9 nm). The atomic lines are detected by a RCA 7102 photomultiplier tube cooled to 240°K (Products for Research), which reduced the dark current by a factor of 10^4 relative to room temperature. The output from the PM tube is amplified by a lock-in amplifier (PAR HR-8) and finally recorded on a strip chart.

The Kr and Xe metastables are produced by adding small amounts of Kr or Xe to the main Ar flow. The following technique was used to optimize the production of Kr* and Xe* but still conserve Xe or Kr. With the Ar lamp in place, Kr (or Xe) (Matheson Gas Products: research grade without further purification) was added to the Ar flow, before the hollow cathode discharge, until $Ar(^3P_2)$ was no longer detectable by absorption at the first observation point. Since the $Ar(^3P_0)$ constitutes ∿13% of the total Ar metastable concentration,[13] and since it reacts with Kr or Xe faster[13] than $Ar(^3P_2)$, removal of $Ar(^3P_2)$ metastables ensures complete removal of the $Ar(^3P_0)$. At this point the Ar lamp was replaced by a Kr (or Xe) lamp, and the Kr (or Xe) flow was increased until no further increase in the absorption of the 811.2 nm Kr line (881.9 nm for Xe) could be seen. Flows of ∿0.6 mmol/min of Kr, as compared to only 0.2 mmol/min of Xe, were required to completely remove all of the Ar metastable atoms. This difference reflects the larger rate constant for quenching of Ar* by Xe.[13]

Absorption of the Ar 811.9 nm line was 20% for a single pass of light; whereas, with Kr or Xe the absorption was only ∿8%. In order to increase absorption for the latter, mirrors were used to obtain triple pass of light through the reactor. Using the procedure outlined in Ref. 13, metastable concentrations of ∿ 5×10^{10} atoms cm^{-3} for Ar* and Xe* and ∿2×10^{10} for Kr* were calculated from the measured absorbances ($\alpha = 3.5$) for several lines terminating in the $^3P_2$ state of each metastable.

Whenever possible, the reagent gases were either distilled or sublimed under vacuum several times before being mixed with argon. The mole fraction of reagent was typically between 0.10 and 0.15; most mixtures were stored in glass reservoirs at a total pressure of 300 torr. Fluorine and other reagents that react with Pyrex glass were stored in a stainless steel cylinder, connected to the calibrated volume (see below) and to the shower-head quencher inlet of the

reactor through 1/4" ss tubing. Although the flow reactor is made of Pyrex, the depletion of F_2 or other fluorine-containing reagents from diffusion to and reaction with the wall is minor at the flow speeds used in this work. This effect, however, would tend to make the measurements lower limits to the true quenching rate constants.

Reagent flows, regulated by a stainless steel fine metering valve (Nupro SS-4SG) were measured by monitoring the pressure increase in a calibrated volume (576 ml) using a calibrated pressure transducer (Celesco Industries PD7), and a chart recorder. Typical reagent flows ranged from 1 to 20 μmoles/min.

Reagent purity is a serious problem for the corrosive chemicals. Extensive gas handling may result in the introduction of as many impurities as are removed. Although manufacturer's specifications usually are quoted to >95% purity, these must be regarded as suspect. Experiments with F_2 were done with mixtures prepared by ourselves from pure F_2 and tank Ar and from a pre-prepared 10% mixture (Matheson). The rate constants from the two mixtures differed by <15%. The rate constants quoted, especially for the very reactive fluorine containing molecules are less reliable than those normally reported from this apparatus ($\pm$ 10% reproducibility).

2B. Measurement of Relative Emission Intensities

Relative emission intensities were obtained by comparing the total integrated excimer emission intensity, corrected for monochromator response, produced by the reaction of $Xe(^3P_2)$ and $Kr(^3P_2)$ with a given halogen donor to the emission intensity produced by the reaction with chlorine. The XeCl and KrCl intensities were then related (see below) to absolute rate constants to obtain branching ratios. The apparatus consisted of a grease-free glass section for handling and storage of non-corrosive gases and a stainless section for handling and storage of corrosive gases. The stainless section contained the pressure transducer (Celesco Industries PD7), which serves the dual purpose of measuring

flows and monitoring pressure in the reaction zone or gas handling section. Metal to glass connections consist of an inner SS standard taper joint machined in our shops and a glass outer standard taper. All metal sections were passivated by filling with pure fluorine at 500 torr for ∿60 hours. In addition, the metal reservoirs were heated to 150°C for 12 hours before the F_2 was removed. The third section sonsisted of the flow meter, metering valve and molecular sieve traps used to regulate and purify the Ar flow. The flow reactor was separated from the mechanical pump (1000 l/min) by a throttling valve (large stopcock) which was partially closed for the high pressure experiments. The Kr and Xe metastable atoms were produced in the usual way (see part 2A) using a hollow cathode discharge between two rolled tantalum foil electrodes (250 V, ∿1mA) located 5.5 cm upstream from the mixing zone.

The reagent gases were the same as those used for the rate constant determination experiments, with the exception of Ar, which was of purity $\geq$ 99.995% to reduce the problem of interfering impurity emissions, N_2(C-B) and OH(A-X). The OH* emission from water impurity was still troublesome for Ar*, and to a lesser extent for Kr*, even with the precautions used here. The fluorine donors were diluted in Ar (mole fractions between 0.25 and 0.45) and stored in the passivated stainless reservoirs. Reagents were added concentrically to the flow of Kr* or Xe* ; the resulting emission formed a diffuse flame some 10 cm in length. All observations were made with the entrance slit of the monochromator parallel to the gas flow and against the spectrosil quartz window, which was 2.5 cm below the origin of the reaction flame. Care was taken to ensure that all of the observations were made in the concentration range where the excimer emission intensity varied linearly with the flow (concentration) of the halogen donor. For this apparatus serious deviation from linearity occurred for reagent flows >5 μmoles. $min.^{-1}$, see Fig. 1, because the metastable atoms were reduced significantly by the high concentration of added reagent (7.4×10^{12} molec. cc^{-3}).

The emission was detected with a 0.75m Jarrell-Ash Czerny-Turner scanning

monochromator equipped with a cooled (0°C) E.M.I. 9558 QB photomultiplier tube and SSR photon counter. The spectral response was calibrated from 300 to 850 nm, using a standard iodine lamp, and by the "molecular branching ratio" method[15,4] from 300 to 190 nm. The resulting calibration curve was fitted piece-wise to five (3 and 4 degree) polynomials using a nonlinear least square computer program. The calibration was checked and confirmed in the 190-300 nm region by using a standard deuterium lamp (Optronic Labs). The counter and monochromater were interfaced to a PDP/8e computer that controlled the scanning and data acquisition. The data were stored on magnetic tapes; these data were corrected for spectral response and reagent concentration and then integrated by simple addition of the counts in each wavelength increment. Normally the scan was taken with increments of 1 Å for low pressure excimer emissions; whereas, high pressure spectra were taken with 0.5 Å increments. In order to test the reproducibility of our results, 5 independent experiments with constant $[Cl_2]$ were performed for $Xe(^3P_2) + Cl_2$ and the total integrated intensity agreed to within 5%. The reproducibility of experiments involving determination of areas of atomic lines was less, ~15%; this seemed to be related to wear in the monochromator screw drive mechanism.

Several of the fluorine donors used in this study are extremely hazardous (with regard to toxicity and as oxidizing agents) and must be handled with great care. In the present work these reagents, after being mixed with the argon, were flowed directly into the pumps on the assumption that reaction with hot pump oil was the best means for disposal. The pumps also were vented to the hoods. With OF_2 small "explosions" in the pumps were noted when evacuating the OF_2/Ar mixture from the storage reservoir.

3. RESULTS

3A. Total Quenching Rate Constants

For Ar the absorption from Pen-Ray lamps can be described[13] by a modified

Beer-Lambert law:

$$I = I_o \exp[-a(k_o \ell)^{\gamma}]; \ \gamma = 0.95 \qquad (1)$$

$$k_o \ell = (2/\Delta\nu_D)(\ln 2/\pi)^{1/2} \ (\pi e^2/mc) \ [M] f \ell$$

Since the experimental conditions are similar for $Kr(^3P_2)$ and $Xe(^3P_2)$, except that the concentrations are smaller than for $Ar(^3P_2)$, and since the source lamps are at the same temperature, eq. (1) should describe the absorption by $Kr(^3P_2)$ and $Xe(^3P_2)$. When a reagent, [Q], is present, the concentration of the metastable atoms [M*] at any point in the reactor can be expressed by[13,16]

$$\ln \frac{[M^*]}{[M^*]_o} = - \frac{D_o}{\Lambda^2 p} + k_1 p + k_2 p^2 + k_Q[Q] \ \frac{z}{v} . \qquad (2)$$

The first three terms arise from the decay processes in argon; the fourth term is the quenching by the added reagent, Q, which is present in sufficiently high concentration that pseudo first order kinetics are appropriate. To obtain k_Q, eqs. 1-2 are combined.

$$\frac{1}{\gamma} \ln\left\{\ln \frac{Io}{I}\right\} \propto \ k_Q[Q] \ \frac{z}{v} \qquad (3)$$

Therefore, for a fixed [Q] plots of $(1/\gamma)$ $\ell n[\ell n \ (Io/I)]$ vs z are linear. Fig. 2 shows these decay curves with distance converted to time for the reaction $Kr(^3P_2) + F_2$. A correction factor of 1.15 arising from converting distance to time is included.[13] Plots of the pseudo first order rate constants vs [Q] give straight lines with slopes, k_Q. Fig. 3 shows plots of K vs [Q] for the data in Fig. 2, and for the reaction of $Kr(^3P_2)$ with NF_3 and $Xe(^3P_2)$ with CF_3OF. Least square fits were used to obtain both pseudo first order rate constants from the decay plots and quenching rate constants from the concentration plots. Typical standard deviations of k_Q were $\pm$ 5%.

Other rate constants were obtained in a similar way and a listing is given in Table 1. The majority of these values were calculated from one experiment at $\sim$0.8 torr; however, tests for reproducibility showed that rate constants

agreed well within < ± 10% for the same mixture. The long term stability of the apparatus was tested by measuring the $Xe(^3P_2)$ + Br_2 rate constant (by different operators) after an interval of 8 months; values of 60 and 63 x 10^{-1} cm^3 $molec^{-1}$ sec^{-1} were obtained. Table 1 includes some reagents previously[13,14] studied with $Ar(^3P_2)$. The large number of rate constants generated patterns which suggested that some reagents needed re-investigation. One problem in particular was that oil manometers were used to monitor flows in some of the previous work[13] which resulted in low values for rate constants of the reagents which dissolved in the oil. The improved $Ar(^3P_2)$ rate constant for Cl_2 (71 vs 47 x 10^{-11} cm^3 $molec^{-1}$ sec^{-1}) is listed in Table 1. Since the Cl_2 reactions with $Kr(^3P_2)$ and $Xe(^3P_2)$ were selected as the reference rate constants, each of these was measured twice; the reproducibility was + 10%. Thus we have good confidence in the Cl_2 results of Table 1. Since the F_2 reactions may be of importance for laser applications, measurements of the rate constants for $Kr(^3P_2)$ were done both with a prepared mixture (same mixture was used for $Ar(^3P_2)$, $Kr(^3P_2)$ and $Xe(^3P_2)$) and with a commercially prepared mixture; the values were 61 and 72 x 10^{-11} cm^3 $molec^{-1}$ sec^{-1} respectively.

Providing that the apparatus is functioning properly, the most likely source of error in rate constant measurements is the preparation of mixtures. The previously published value for HBr with $Ar(^3P_2)$ seems to be a case in point. Re-investigation gave 52 x 10^{-11}, rather than the previous value of 77 x 10^{-11} cm^3 $molec^{-1}$ sec^{-1}. In the current work care was taken to prepare pure HBr free from H_2 or Br_2 contamination. With the exception of reagents with slow quenching rates, the rate constant values for a given reagent with all three metastables tend to be very similar. When transcribed into thermal quenching cross sections, a virtually constant ratio is obtained since the mean speed is less for the heavier rare gases.

3B. Relative KrF and XeF Excimer Emission Intensities.

Collisional quenching of rare gas metastable atoms by molecules has many exit channels, i.e., Penning ionization, associative ionization, excitation transfer, dissociation or dissociative excitation, and most notably, chemical reaction with the production of (MX)* + R:

$$\begin{aligned} M^* + RX &\xrightarrow{k_{PI}} M + RX^+ + e^- \\ &\xrightarrow{k_{AI}} (MRX)^+ + e^- \\ &\xrightarrow{k_E} M + RX^* \\ &\xrightarrow{k_D} M + R + X \\ &\xrightarrow{k_{DE}} M + R^* + X \text{ (or, } M + R + X^*) \\ &\xrightarrow{k_{MX}} (MX)^* + R \xrightarrow{\tau^{-1}} M + R + h\nu \end{aligned} \tag{4}$$

The "Branching Ratios" can be defined as:

$$BR_i \equiv \frac{k_i}{\sum_j k_j} = \frac{\sigma_i}{\sum_j \sigma_j} \tag{5}$$

where the k_i's and σ_i's are the rate constants (or cross sections) for the production of the given product, and $\sum_j k_j$ and $\sum_j \sigma_j$ are the total quenching rate constant (k_Q) and cross section (σ_Q) of Table 1. For Xe* and Kr* with fluorine donors, ionization is not important but is included here for generality. The main interest of the current work is the branching ratio for the formation of the rare-gas fluoride excimer, k_{MX}/k_Q. Instead of attempting direct measurement of k_{MX}, a comparison technique was developed.

The rate of formation of MX*,

$$M^* + RX \xrightarrow{k_{MX}} R + MX^* \xrightarrow{\tau^{-1}} R + M + X + h\nu \ , \tag{6}$$

is given by (7),

$$I_{MX}^{RX} = k_{MX}[M^*][RX] \ , \tag{7}$$

provided there is no quenching or non-radiative decay (such as predissociation) of MX. Therefore, if k_{MX} is assigned for a reference reaction, we can obtain others simply by comparing the total relative emission intensities.

$$\frac{I_{MX}^{R'X}}{I_{MX}^{RX}} = \frac{k'_{MX}[M^*][R'X]}{k_{MX}[M^*][RX]} \tag{8}$$

If experiments are done for the same [M*], we obtain

$$k'_{MX} = k_{MX} \frac{I_{MX}^{R'X}/[R'X]}{I_{MX}^{RX}/[RX]} \; ; \; \sigma'_{MX} = \left(\frac{\mu'}{\mu}\right)^{1/2} \sigma_{MX} \frac{I_{MX}^{R'X}/[R'X]}{I_{MX}^{RX}/[RX]} \tag{9}$$

μ and μ' are the reduced masses for the collision pairs M* + RX and M* + R'X, respectively.

Molecular chlorine was selected as the reference RX because the absolute values for k_{XeCl} and k_{KrCl} (or σ_{XeCl} and σ_{KrCl}) could be estimated, as will be shown in section 3C. A second reason for selecting Cl_2 was that it can be stored and metered in a reliable way, thus making the determination of concentrations more straight forward. The XeCl and KrCl emission spectra are shown in Figures 4, 5 and 6. The band on top of the KrCl secondary emission is the Cl_2^* 256 nm emission which also is produced from Ar* + Cl_2 (Fig. 4). The Cl_2^* emission, which corresponds to 10% of the total KrCl emission intensity, was subtracted when comparing the KrCl* emission intensity with that of KrF* from RF reactions. One of the less desirable aspects of using KrCl* as the reference is that the emission extends to short wavelengths where the monochromator calibration is less reliable. About 10% of the KrCl emission was below the λ range observable with the Jarrell Ash monochromator; this correction to I_{KrCl} was determined from a spectrum taken with a vacuum instrument.

Since variation of pressure above ∿2 torr has a significant effect upon the appearance of the excimer emission intensities, all comparisons were made in the 1 - 2 torr pressure range. We observed no quenching of the main excimer emission (see Discussion) up to 40 torr of argon; however, the appearance of the spectra changes markedly with increasing pressure as demonstrated in Figures 5 and 6. These effects, which have been discussed previously for ArCl,[4] can be largely explained by vibrational relaxation of the excimer molecules initially formed in very high vibrational levels. However, since the secondary emission also is reduced relative to the primary band, vibrational relaxation probably is not the whole explanation and collisional transfer between two ArCl electronic states ($^2\Sigma^+$ and $^2\Pi_{3/2}$) may be responsible for a changing primary/secondary emission. The secondary emission is very weak for XeF and KrF; however, for the chlorides the reduction in the secondary emission is clearly observable and both electronic and vibrational relaxation seem to be occurring. The presence of a third excimer state ($^2\Pi_{1/2}$) can be noted by the small, sharp band appearing in the spectra of Fig. 5-8. Further consideration of this state and the changes of the spectra with pressure will be given in the Discussion Section. In obtaining the relative emission intensities, the integration included the main band and the secondary and excited state contributions; the latter actually makes a negligible contribution for all reagents listed in Table II except XeF_2.

Table II lists the relative rate constants (or relative emission intensities for equal concentrations of RX) for some F-donors grouped according to the location in the Periodic Table of the element bound to F. Some representative XeF and KrF spectra are shown in Figures 7-9. Also listed in Table II are the bond energies of the F-R donors and λ_{min}, the observed short wavelength limit (at low pressure) for onset of the XeF emission. The short wavelength limits of the KrF emission are not given because the declining response of our detection system makes these difficult to assign. The relative intensities listed in Table II are the averages of at least two scans performed during the same

experiment, and in several cases (Groups VII & V) the average value includes experiments done weeks and even months apart. Immediately before and/or after each M* + RX experiment a M* + Cl_2 experiment was run. For the same gas mixture, the uncertainty of the relative rate constants is ± 10%. In a given experiment, the main source of error seems to have been the instability of the discharge and the consequent variation in [M*] over the 3 to 4-hour period required to obtain the scan for Cl_2 and RF. For the highly reactive fluoride donors, sample preparation and storage are the greatest problems.

Except for NOF, which gave NO(A) and NO(B), no other emissions were observed over the 800 - 200 nm region from RF reactions. Although all R-F molecules quench metastable atoms very readily, dissociative excitation is quite rare with $Kr(^3P_2)$ and $Xe(^3P_2)$ reactions. Since ionization is energetically forbidden, the quenching which does not yield MF must involve dissociation with excitation. Similar results were found for $Ar(^3P_{0,2})$ reacting with chlorine donors.[4]

3C. Absolute Determination of k_{MCl}

In order to put the relative rate constants of Table II on an absolute basis, the absolute value of the rate constant for formation of MCl from the reaction of Cl_2 with $Xe(^3P_2)$ and $Kr(^3P_2)$ must be assigned. Since the absolute rate constants for the formation of excited Kr states by the reaction of $Ar(^3P_2)$ with Kr are known,[23] Kr can be used as an internal standard in the following way. A mixture of Cl_2 and Kr (normally 50-50) was reacted with $Ar(^3P_{2,0})$ and the relative emission intensities of the Cl_2* 256 nm continuum and the Kr* 760.2 nm line were compared. Using the reported value[23] for the rate constant for formation of the Kr $[5p(3/2)_2]$, which is the source of the 760.2 emission, the rate constant for formation of the Cl_2* 256 nm continuum can be calculated. The average of 35 determinations with three mixtures gave $(1.7 \pm 0.2) \times 10^{-11}$ cm^3 $molec^{-1}$ sec^{-1} for k_{Cl_2}*. At this point, the concentration of Ar* was measured using absorption spectroscopy and then a flow of

Kr was added to form $Kr(^3P_2)$. The concentration of $Kr(^3P_2)$ was measured using absorption spectroscopy, see Table 3, and finally this Kr* concentration was reacted with the same flow of Cl_2 as used with $Ar(^3P_{0,2})$. The spectra from Ar* + Cl_2 and Xe* + Cl_2, using this procedure, are shown in Fig. 4. In making this comparison only the ratio of the $Ar(^3P_{0,2})$ and $Kr(^3P_2)$ (or $Xe(^3P_2)$) concentrations are needed, e.g.,

$$\frac{I_{Cl_2^*}}{I_{KrCl}} = \frac{k^{Ar}_{Cl_2}}{k_{KrCl}} \frac{[Cl_2]}{[Cl_2]} \frac{[Ar(^3P_{0,2})]}{[Kr(^3P_2)]} \quad . \tag{10}$$

Both $Ar(^3P_2)$ and $Ar(^3P_0)$, which are present in a ~6:1 ratio, are assumed to give Cl_2^*. Combining the relative intensities of Cl_2^* 256 nm continuum and the KrCl emission and the measured Ar* and Kr* concentrations of Table 3 gave $k_{KrCl} = 52 \times 10^{-11}$ cm^3 molec^{-1} sec^{-1}. Some words of caution, however, are in order. The comparison of the Cl_2^* 256 nm continuum and the Kr* 760.2 nm line are subject to some uncertainties because the comparison requires the use of the ends of our calibrated spectral response curve, which could be in error by $\pm$ 20%. Secondly the atomic transition probabilities are not reliably known as reflected in the variation of the Ar* and Kr* concentrations for use of different lines as shown in Table 3. However, the mean value for the $Kr[^3P_2]$ concentration is fully consistent with expectations based upon formation of $Kr(^3P_2)$ by the $Ar(^3P_2)$ + Kr transfer reaction[23] followed by radiative branching to $Kr(5s,^3P_2)$ and $Kr(5s,^3P_1)$. The experiment described below suggest little direct formation of $Kr(^3P_2)$ in the discharge; therefore, we have confidence in the measured $[Kr(^3P_2)]/[Ar(^3P_2)]$ ratio. This value for k_{KrCl} when corrected for the emission below λ = 190 nm gives a branching ratio for KrCl* and Cl_2^* formation of 0.70 and 0.07, respectively.

Application of the above procedure to $Xe(^3P_2)$ + Cl_2 is more difficult, since there are no reliable values for the transition probabilities of the $Xe(6s^3P_2)$ - Xe*(6p) transitions. For some other transitions calculated values

disagree with experimental measurements by as much as 100%. For example, Allen et. al.[22] measured the lifetimes of the Xe $6p[1/2]_0$ and $6p[3/2]_2$ states (to our knowledge the only measurement on the 6p-6s array of Xe(I)) and obtained (40±12) and (33±20) ns, respectively. The calculated transition probabilities of Chen and Gastang[23] give reasonable agreement for the $6p[3/2]_2$ state, 29 ns; however, the calculated transition probability for just the $6p[1/2]_0 - {}^3P_1$ transition gives 28 ns for the lifetime of the $6p[1/2]_0$ state. If the calculated values of Chen and Gastang are used, the concentrations shown in Table 3 are obtained. According to these results, the concentrations of $Kr({}^3P_2)$ and $Xe({}^3P_2)$ were approximately the same. As will be shown below, this is incorrect and, in fact, $[Xe{}^3P_2]/[Kr,{}^3P_2] \tilde{\sim} 2.5 - 3$. Therefore the calculated transition probabilities for Xe* require revision.

The transfer reaction of $Ar({}^3P_2)$ with Xe gives a large variety of Xe* states[27] and calculating a reliable ratio for the total radiative cascade to the metastable and resonance Xe(6s) states is not feasible. Furthermore, direct excitation of $Xe({}^3P_2)$ occurs in the discharge as well as formation via radiative cascade from the Ar* + Xe excitation transfer reaction. This was shown as a consequence of further effort to evaluate the $[Xe, {}^3P_2]$. An experiment was done in which the XeCl and KrCl and the XeF and KrF emission intensities were compared when $Xe({}^3P_2)$ and $Kr({}^3P_2)$ were made from the same Ar* concentration and with the same Cl_2 and F_2 concentrations. The emission intensities of XeF and XeCl were <u>threefold</u> larger than for KrCl and KrF. Since the branching ratios for the $Kr({}^3P_2)$ reactions already have been shown to be ∿ unity, $[Xe,{}^3P_2]$ must be greater than $[Kr, {}^3P_2]$. Since the $Ar({}^3P_2)$ transfer reaction cannot have unit conversion to $Xe({}^3P_2)$, the implication is that $Xe({}^3P_2)$ is directly produced in the discharge. This implication was confirmed by comparing the XeCl emission intensities when xenon was added before and after the discharge. Enough Xe was added to ensure that the excitation transfer was completed before the Xe/Ar mixture reached reaction zone.

The emission intensity for addition before the discharge was 2.4 times larger than for Xe addition after the discharge, which proves quite conclusively that direct production of $Xe(^3P_2)$ occurs in the discharge. A similar experiment with Kr gave $\leq$ 15% increase in $Kr(^3P_2)$ concentration for addition before the discharge. We, thus, conclude that the higher XeCl and XeF emission intensities, relative to KrCl and KrF, can be attributed to the larger $Xe(^3P_2)$ concentrations and that the branching ration for XeCl and XeF formation from Cl_2 and F_2 must be very close to unity. Although the $Kr(^3P_2) + F_2$ branching ratio was measured as 0.7, recalibration measurements from 250-190 nm (which are in progress with the D_2 lamp) strongly suggest that the final ratios between the KrF and KrCl intensities will lead to unity branching ratio for KrF formation from F_2. This anticipated result was used in Table 5. The final branching ratios are tabulated in Table 5 for the better fluorine donors. Although not included in the table, the branching ratio for the $Xe(^3P_2) + XeF_2$ reaction are thought to be large.

4. Discussion

4A. Rare Gas Excimer State for Krypton and Xenon Halides

The states of the excimers will be discussed by using the recent calculations[28] for KrF as reference. As shown in Figure 10, there are three states that correlate with $Kr(^1S_0) + F(^2P_{3/2})$ and $F(^2P_{1/2})$ and three states that correlate with $F^-(^1S_0) + Kr^+(^2P_{3/2})$ and $Kr^+(^2P_{1/2})$. The latter are basically coulombic states and resemble the alkali metal-halide ground states. A fourth bound state in the 9-10 eV range, with smaller R_e, correlates with $F(^2P_{3/2})$ and $Kr(^3P_2)$ and is non-coumbic in nature; this state does not play a role in the present discussion but is important in quenching of metastable rare gas atoms by atomic halogen.[27] According to Dunning and Hay's calculations, the lowest excimer state is the $^2\Sigma^+$ (or $\Omega = 1/2$) state. However, the calculated separation between the $^2\Pi_{3/2}$ (or $\Omega = 3/2$) and the $^2\Sigma^+$ states is very small for KrF and there still may be some question about the ordering for other rare gas halides. The main rare gas halogen excimer emission is usually assigned as $(^2\Sigma^+ - {}^2\Sigma^+)$; however, for the lower pressure spectra this transition could be

emission from $^2\Pi_{3/2} - {}^2\Sigma^+$. The secondary emission (that portion of the emission extending to the red of the strongest band) is understood less well but probably involves emission from $^2\Pi_{3/2}$ to $^2\Pi_{1/2,3/2}$ with possibly some contribution from

$^2\Sigma^+$ to $^2\Pi_{1/2,3/2}$. The secondary emission is quite pressure dependent, vide infra. Both the main and secondary emissions generated from reactive quenching at low-pressure show interference patterns[5] characteristic of bound-free transitions. As the pressure is increased, vibrational relaxation occurs, and for sufficiently high pressure the XeF and XeCl emissions become bound-bound; these spectra have been assigned and spectroscopic constants obtained.[8,18] For some of the rare gas halides, especially ArCl, ArBr and ArI, the bound coulombic states are very likely predissociated by diabiatic states correlating to rare gas + excited halogen states.[4] This is not of importance for Kr or Xe with F or Cl.

The main contribution of the present work to further elucidation of the excimer states is the identification of the $^2\Pi_{1/2}$ ($\Omega = 1/2$) - $^2\Sigma^+$ emission. A definitive spectrum (see Figure 9) was obtained from the reaction of $Xe(^3P_2)$ + XeF_2; the appearance of the 260 nm band is very similar to that of the main emission at 350 nm and the 260 nm band clearly can be assigned to the $^2\Pi_{1/2}$ - $^2\Sigma^+$ transition as previously suggested by Ewing and Brau.[6] After obtaining the 260 nm XeF* spectrum from XeF_2, it was possible to identify this transition, which normally is very weak, in spectra from many other halogen donors with low bond energies. The results for Xe with all halogens and Kr with Cl and F are tabulated in Table 5. The energy separation between the $^2\Sigma^+$ and $^2\Pi_{1/2}$ excimer states was calculated from the peaks of the maximum intensity bands, or from the sharp 0-0 bands of XeF. As shown in Table 5, this energy difference is nearly constant for the various halogens. Furthermore, the energy separation is nearly the same as the asympotic separation, as might be expected since the bonding is dominated by coulombic forces. The energy difference is actually slightly less than the asympotic separation so there is less repulsion in the $^2\Sigma^+$ than in the $^2\Pi_{1/2}$ state. Since these two states have nearly the same separation at R_e as in the asympotic ion pair limit, this implies that the

separation between the $^2\Pi_{3/2}$ and $^2\Sigma^+_{1/2}$, which correlate to a common limit, should be quite small, as also shown by the calculated results of Figure 10.

The secondary emission, which is thought to represent termination on the $^2\Pi$, rather than the $^2\Sigma^+$ lower state, is very weak for XeF and KrF. Therefore, attention can be focused on just the variation of the main emission with pressure. As shown in Figure 8, the XeF emission intensity shifts toward the long wavelength limit with increasing pressure and acquires structure characteristic of a bound-bound emission originating from the lower vibrational levels of the upper state. Since emission from high vibrational levels of XeF is bound-free, the spectrum from a broad distribution of levels consists of both bound-free and bound-bound contributions. Thus vibrational relaxation occurred without any significant change in the relative integrated intensity of the main XeF* and NO(A-X) emissions. Since the NO(A) state is not quenched by argon, evidently more than 40 torr of Ar is required for quenching of XeF*. The observation of extensive vibrational relaxation over the pressure range from of 1-10 torr implies that the radiative lifetime is $\geq$ 50 nsec, since the time between gas kinetic collisions is $\sim$50 nsec at 5 torr. A more precise estimate will require computer modeling of the change in the emission spectrum vs pressure. This estimate is about a factor of 2 larger than the value obtained for XeBr by Searles and Hart[9] from pulsed e-beam excitation.

The most perplexing problem is the explanation of the secondary emission, which has greater intensity for combinations of the heavier halogens and lighter rare gases. The secondary KrCl emission, which has the Cl_2* 256 nm continuum superimposed upon it, is clearly evident in Figure 5. As the pressure is increased, the secondary emission declines and shifts to the blue and at the highest pressure the Cl_2* band has become dominant. The decline of the secondary emission intensity relative to primary emission with pressure previously has been noted with ArCl*.[4] Also, the ratio of the main and secondary ArCl* emission

intensities varied with donor. It was concluded that two ArCl* electronic states must be involved and that collisional transfer between electronic states as well as vibrational relaxation must occur at increased pressure. The present work confirms this view but provides no further insight except to emphasize that the secondary emission and its variation with pressure are not well understood. Since the $Xe(^3P_2)$ and $Kr(^3P_2)$ reactions with RF are not expected to be any more specific than with RCl with respect to formation of given excimer states, the radiative branching ratios must change for the fluorides vs chlorides. Thus, the question becomes, why are the radiative transitions terminating on the $^2\Pi_{1/2,3/2}$ states less favorable for the fluorides than for the chlorides.

Ewing and Brau[7] reported a broad smooth emission system near 450 nm for e-beam excitation of Xe and F_2 mixtures. For XeF formed by direct reaction, the XeF secondary emission is observed as a very long but flat tail extending from 360 to 500 nm. This secondary emission does not appear to be the same as that observed by Ewing and Brau. The third $^2\Sigma^+$ at ~9 eV, which correlates to Kr* + F, is crossed by the coulombic curves at short range. If the avoided crossing is sufficiently strong a bound state might result. However, the only evidence to date is the study[31] of $Ar(^3P_2)$ + Cl, which suggests that the upper $^2\Sigma^+$ curve is pre-dissociated. Thus it seems unlikely that emission from the upper $^2\Sigma^+$ potential will be observed.

4B. Reactive Quenching of Rare Gas Metastable Atoms by RF.

Efficient formation of XeF and KrF evidently only occurs for a limited choice of fluoride donors, e.g. halogens and small molecules with the OF or NF bond. Detailed consideration (e.g. arguments of electron affinity, nature of the lowest unfilled orbital, stability of RF^-, etc.) of the reaction mechanism for reactive quenching with formation of KrF or XeF will be delayed until after the other halide donors have been studied. At the present time only a general summary will be attempted with emphasis on the decline in the branching ratio

as RF becomes larger. The general interaction mechanism with most RF molecules appears to be a charge transfer mechanism in which the entrance channel, $V(M^*,RF)$, is crossed at relatively long range by an ion pair potential, $V(M^+, RF^-)$. Efficient transfer to $V(M^+,RF^-)$ is followed by a competition between formation of MF* + R and M + RF*. The RF* channel is observed as either molecular fluorescence, e.g. Cl_2^*, or as dissociation with excitation of R*. For most of the RF systems dissociation occurs without excitation of R*; however, there are other systems (such as RCN)[26] that give a large amount of dissociative excitation. For diatomic halogens one expects the reactions to be direct. This is born out by the formation of MX* in high vibrational levels with emission arising from levels corresponding to the thermochemical limit for Cl_2 and F_2. However, for larger RF some evidence suggests that M^+RF^- can be longer lived. Excimer formation occurred from the central atom with BrF_5 and IF_5, and some time presumably is required for this rearrangement. Furthermore, the XeF*($^2\Sigma$ or $^2\Pi_{3/2}$) product formed from XeF_2 was more vibrationally relaxed than from N_2F_4, which has about the same bond energy, suggesting a lifetime sufficiently long for some dissipation of energy into internal modes of the $Xe^+-XeF_2^-$ complex. Thus the reaction dynamics for metastable rare gas atoms + RF probably span the scale from being direct, to cases involving longer lived complexes for larger RF molecules with higher $D°(R-F)$ values[30]. With this qualitative view, it is not surprising that the branching ratio for excimer formation only approaches unity for small molecules with low bond energies. Based upon the formation of $Cl_2^*(E)$ from the $Ar(^3P_2)$ and $Kr(^3P_2)$ + Cl_2 reactions, escape from the charge transfer potential, $V(M^+, RF^-)$ to M + RF* can be quite rapid. Thus if the charge transfer complex has an increased lifetime with respect to entering the MF* + R channel, the M + RF* channel seems to dominate the exit channel distribution. Presumably the probability for the latter also increases with larger R because of the greater number of excited state RF* potentials. An alternative explanation is that the transfer efficiency from V(M*, RF) to

$V(M^+, RF)$ declines with increasing size of RF; quenching then occurs by direct transfer from V(M*, RF) to V(M, RF*) at shorter range. Although we do not generally favor this explanation, it may be of importance for some reagents.

Reactions known to occur via the direct long-range ionic curve crossing mechanism release a large fraction of the available energy to vibrational energy of the halide product.[31] Therefore for the metastable atom reactions proceeding by this mechanism the MX molecules should be formed in high vibrational levels and the emission should extend to the thermochemical limit. This certainly is true for $Xe(^3P_2)$ + Cl_2 and F_2. Within the uncertainty of the bond energies and the assignment of λ_{min}^{obs} (the total absence of $Ar(^3P_{0,2})$ must be assured since the ArF secondary emission overlaps the tail of the main XeF emission), the XeF emission from the FNO, OF_2 and CF_3OF reactions also extend to the thermochemical limit. The situation for NF_3 and N_2F_4 is more complicated. For N_2F_4 the emission extends to smaller λ than expected for a bond energy of 71 kcal $mole^{-1}$ and these data suggest $D(F\text{-}N_2F_3)$ is less than 71 kcal $mole^{-1}$. There is general agreement that $D(F\text{-}NF_2)$ is $\sim$57 kcal $mole^{-1}$, nevertheless the emission clearly does not extend to the thermochemical limit. Since XeF emission from N_2F_4 does extend to the thermochemical limit, it seems unlikely that either the N_2F_4 or NF_3 reactions involve long lived complexes. One possibility is that NF_2 is a stablized radical and that the stablization energy is not available[32] to the XeF product. The Xe* + XeF_2 reaction merits special consideration. The reaction obviously yields $XeF(^2\Sigma^+$ or $^2\Pi_{3/2})$ in very low vibrational levels simultaneously with formation of $XeF(^2\Pi_{1/2})$ and some $XeF(^2\Sigma^+$ or $^2\Pi_{3/2})$ in high vibrational levels. In fact the emission does extend to the thermochemical limit (conditional upon the present assignment of D(F-XeF) being correct). It appears that there may be two distinct components which may be associated with the break-up of the Xe^+, Xe^-F_2 complex. We tentatively suggest that these components can be associated with different Xe atoms in the complex. More precise deductions must await computer modeling of the spectra to establish vibrational populations.

The total reactive cross sections of the metastables with halogens (at 300°K) can be compared with those of the alkali metals[33] (at ∿ 1000°K). The values for Cl_2 with K(Ar*), Rb(Kr*) and Cs(Xe*) are 154 (142), 190 (179) and 196 (193), $\mathring{A}^2$, respectively. For Br_2 the comparison gives 151 (147), 197 (?) and 204 (202) $\mathring{A}^2$. The similarity is quite remarkable. The variation of the cross sections with temperature is small[31] and will not significantly affect these comparisons. Based upon an optical model, the differential scattering data of $Ar(^3P_{0,2})$ with HBr have been analyzed[34] to obtain total inelastic cross sections at four energies ranging from 60-160 meV. The thermally averaged value from this study gives 14.7 $\mathring{A}^2$. Although our previously published value was about 20% too high, the cross section in Table *I* is still ∿8 times larger than the results from the beam work.[34] The total quenching cross sections for all of the hydrogen halides exceed 100 $\mathring{A}^2$, and the magnitude of the inelastic cross section deduced from the beam measurements must be questioned.

In earlier work[13] a linear correlation of total quenching cross sections, σ_Q, vs C_6, the long range attractive van der Waals coefficient, was found for a variety of reagents but that selection of reagents did not include many halogen containing molecules. A similar plot is shown in Figure 11 for the quenching of $Ar^*(^3P_2)$ by halogen containing molecules. The C_6 parameters were calculated using the Slater-Kirkwood approximation; dipole induced terms were omitted. The polarization for molecules used in the calculations were obtained from the literature[35,36] or were estimated using additive values given by Applequist, Carl and Fung.[36] The effective number of electrons used in the Slater-Kirkwood approximation was taken to be the sum of the valence shell electrons of the elements making up the molecule in question. The C_6 parameters calculated in this manner will have an approximate uncertainty of $\pm$ 20%. Somewhat surprisingly the correlation plot for the halogens still is remarkably good and, since there is nearly a constant ratio between the cross sections for

$Ar(^3P_2)$, $Kr(^3P_2)$ and $Xe(^3P_2)$, a similar correlation plot also holds for the quenching of $Kr^*(^3P_2)$ and $Xe^*(^3P_2)$. Notable exceptions occur for quenching by F_2 and to a lesser extent with OF_2. The large cross sections of F_2 and OF_2 in spite of their low polarizability can be explained in terms of direct reaction via the long range curve crossing mechanism between the input channel and the charge transfer potential. The more puzzling question is why the other halogens follow this C_6 correlation so closely when we have just argued that quenching occurs by the charge-transfer curve-crossing mechanism. A tentative explanation, pending detailed investigation, is that the inclusion of polarizability into the curve crossing model (which evolves to an orbiting collision model)[33] will have the effect noted in Figure 11.

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Table 1. Quenching Rate Constants (10^{-11} cm^3 $molec^{-1}$ sec^{-1})

Reagent	Ar^3P_2		Kr^3P_2		Xe^3P_2	
	k_Q	$\sigma(\AA^2)$	k_Q	$\sigma(\AA^2)$	k_Q	$\sigma(\AA^2)$
Cl_2	71[b]	142	73	179	72	193
F_2	75	132	72	146	75	161
Br_2	65	147			60	202
ICl	61	138			50	171
OF_2	57	107	53	121	57	139
CF_3OF	43	91	42	114	47	143
SF_6	16[a]	36			28[a]	94
SeF_6	71	166			65	246
TeF_6	58	135			63	230
SO_2F_2	42	89				
NF_3	14	28	16[c]	39	9[c]	23
N_2F_4	31	65	33	90		
NOF	36[d]	68	47	102	39[d]	93
NOCl	48	95			51	135
HCl	37[a,b]	65			56	119
HBr	52[b]	106			61	173
HI	75[a]	166				
CF_3I					52	184
BrCN	46	100				
PCl_3	53	116				

a. Taken from previous work, Ref. 13 and 14.

b. See text for comment relative to Ref. 14.

c. The rate constant measurements for NF_3 were done with the same mixture; the value for $Xe(^3P_2)$ was confirmed with a new mixture. Inspection of the $Kr(^3P_2)$ data suggests that the least square value for this rate constant may be somewhat high, but lack of sample prevented a check for $Kr(^3P_2)$.

d. The rate constants for NOF with $Ar(^3P_2)$ and $Xe(^3P_2)$ may be somehwat low because the NOF mixture was stored in glass. The rate constant with $Kr(^3P_2)$ was measured using the ss. reservoir. Based upon the general trends in this table the rate constants for $Ar(^3P_2)$ and $Xe(^3P_2)$ would not be expected to be larger than 50 x 10^{11} cm^3 $molec^{-1}$ sec^{-1}.

Table 2 - RELATIVE RATE CONSTANTS FOR EXCIMER FORMATION

Group	Reagent	D_o^o [a] kcal/mole	$k_{(Xe)}$	$k_{(Kr)}$	$\lambda^{obs}_{min}-\lambda^{c}_{min}$ nm [b]
	Cl_2	57.9	100	100	214-214
VIII	XeF_2	61[c]	Strong[d]	Strong[d]	217-219
VII	F_2	38	100	100	190-186
	ClF_3	54[e]	33[f]	63[f]	207-
	BrF_5		0,(23[g])	0,(33[g])	193-
	IF_5		0,(~5[g])		205-
VI	OF_2	40[i]	70	82	202-188
	CF_3OF	44[h]	25	35	205-194
	SF_6	>~78	~0	~0	
	SOF_2	~86[i]	~1	~1	308-271
	SO_2F_2	~84[j]	0	0	
	SF_4	~83[k]	0	0	
	SeF_6		0	0	
	TeF_6		0	0	
V	NF_3	57	14	7	245-212
	N_2F_4	71[l]	27	22	230-237
	ONF	55	40,(43)[m]	<5[n]	220-209
	PF_3		<1.0		
IV	CF_4	125	0	0	
	SiF_4	110	0	0	
III	BF_3	134	0	0	

[a]Unless indicated otherwise, the bond energies were taken from reference 17.

[b]λ_{min}^{obs} and λ_{min}^{c} are the observed and calculated $(E(Xe,{}^3P_2) - (D(R\text{-}F))$ short wavelength limit of the XeX emission spectra.

[c]Calculated from $\Delta H_f^\circ(XeF_2)$ assuming $D_o^\circ(XeF) = 2$ kcal mole^{-1} (see reference 18).

[d]Since the XeF_2 flows were not measured accurately, relative rate constants can be not be assigned.

[e]Calculated from the observed $\lambda_{min}(XeF)$; therefore this value is an upper limit.

[f]This relative rate constant represents the total fluoride and chloride excimer emission; the approximate ratio of I_{XeF}/I_{XeCl} was 1.5 (see Fig. 7).

[g]Only XeBr, KrBr or XeI emission was observed.

[h]Ref. 19.

[i]Average of the values reported in ref. 20.

[j]Average bond energy calculated from thermochemical data.

[k]See ref. 22.

[l]Ref. 21

[m]Includes emission from NO(γ) bands, see Fig. 10.

[n]The NO(γ) and NO(γ) bands are present and constitute most of the emission; the listed value corresponds to the estimated KrF emission intensity.

Table 3 - Metastable Atom Concentrations from Atomic Absorption

M	Transition	λ(nm)	A	$k_o\ell$[d]	f		[M*][e]
Ar	$1s_5 - 2p_9$	811.5	0.1685	0.802	0.380[a]	5.1	(E10)
	$1s_5 - 2p_6$	763.5	0.0855	0.354	0.223[a]	4.1	(E10)
Kr	$1s_5 - 2p_6$	760.2	0.0629	0.250	0.2575[b]	1.7	(E10)
	$1s_5 - 2p_7$	810.4	0.0283	0.106	0.1026[b]	1.7	(E10)
	$1s_5 - 2p_9$	811.3	0.1257	0.550	0.4839[b]	1.9	(E10)
Ar	$1s_5 - 2p_9$	811.5	0.193	0.965	0.380[a]	6.1	(E10)
	$1s_5 - 2p_8$	801.5	0.0363	0.140	0.105[a]	3.3	(E10)
	$1s_5 - 2p_6$	763.5	0.100	0.420	0.223[a]	4.8	(E10)
Xe	$1s_5 - 2p_8$	881.9	0.1178	0.510	0.4897[c]	1.3	(E10)
	$1s_5 - 2p_6$	823.2	0.0521	0.203	0.234[c]	1.1	(E10)
Ar	$1s_5 - 2p_9$	811.5	0.1869	0.922	0.380[a]	5.9	(E10)
	$1s_5 - 2p_8$	801.5	0.0490	0.190	0.105[a]	4.4	(E10)
	$1s_5 - 2p_6$	763.5	0.1048	0.460	0.223[a]	5.3	(E10)
Xe	$1s_5 - 2p_8$	881.9	0.1210	0.528	0.4897[c]	1.3	(E10)
	$1s_5 - 2p_6$	823.2	0.0618	0.248	0.234[c]	1.4	(E10)

[a] See Ref. 13; the $^1S_5 - ^2P_8$ transition was too weak to be reliably observed for the first set of data.

[b] Calculated from transition probabilities of reference 24 using the relationship $f_{ik} = (A_{ik})\ 1.4992\ \lambda^2\ (j_k/g_i)10^{-16}$.

[c] Calculated from transition probabilities of Ref. 26.

[d] $k_o\ell$ was obtained from a prepared plot of A_α vs $\sum_n (-1)^n \frac{1}{n!} \frac{(k_o\ell)^n}{(1+n\alpha^2)^{1/2}}$

[e] $[M^*] = 7.1461\ E5\ (\frac{T}{M})^{1/2}(\frac{k_o\ell}{f\lambda})$.

Table 4 - Branching Ratios for KrF and XeF Formation

Reagent	$Xe(^3P_2)$	$Kr(^3P_2)$
F_2	1.0	1.0
OF_2	0.92	1.11
CF_3OF	0.40	0.60
NF_3	1.16	$\geq$ 0.31[c]
N_2F_4	0.63[b]	0.48
NOF	0.77	0.08

[a] The uncertainities are estimated as $\pm$ 0.20.

[b] The k_Q value was taken to be the mean of the rate constants for $Ar(^3P_2)$ and $kr(^3P_2)$.

[c] The quenching rate constant for NF_3 is thought to be large; hence, the branching ratio may be too low.

Table 5. The $^2\pi_{1/2} - {}^2\Sigma^+$ Excimer Transition

Molecule	Wavelength(nm)[a]	Energy(eV)	Separation[b] from $^2\Sigma^+$(eV)
XeF	264.2	4.702	1.16
XeCl	235.8	5.269	1.21
XeBr	221.3	5.615	1.20
XeI	203.0	6.123	1.21
KrF	220.5	5.636	0.62
KrCl	198.7	6.255	0.65

a. For XeF, this wavelength was measured at the sharp band that resembles the 351.1 nm band of the main emission. For all others, the wavelength corresponds to the maximum of the peak.

b. The asympotic separation of the $^2P_{1/2}$ and $^2P_{3/2}$ states of Xe^+ and Kr^+ are 1.307 and 0.665 eV, respectively. The band separation was calculated from the 0-0 laser transitions, except for XeI. In this latter case, we used 253 nm as the head of the main emission.

FIGURE CAPTIONS

Figure 1 - Excimer emission intensity as a function of added Cl_2 flow: ○ -XeCl*, monochromator slits = 150 microns; □-KrCl* monochromator slits = 120 microns. For the experiments reported in this work, quencher flows never exceeded 3 μmoles min^{-1}. A flow of 5 μmoles min^{-1} corresponds to 6.3×10^{12} molec. cm^{-3}.

Figure 2 - "Decay" plots of $Kr(^3P_2)$ for various F_2 concentrations (moles cm^{-1}); ○-0.98×10^{11}; □-2.17×10^{11}; ▲-2.90×10^{11}; ⬡-4.23×10^{11}; ▽-5.98×10^{11}. Total pressure was 0.9 torr.

Figure 3 - Pseudo first order rate constants vs [Q]. Triangles, $Kr(^3P_2)$ + F_2; squares, $Kr(^3P_2)$ + NF_3; circles, $Xe(^3P_2)$ + CF_3OF. The lines are least squares fits to the points. Open circles with a central + are calculated points from least square line.

Figure 4 - Emission spectra from $Ar^*(^3P_{0,2})$ + Cl_2 (upper) and $Xe(^3P_0)$ + Cl_2 (lower) at 1.5 torr; the different scales of the photon counter should be noted. For both cases the Cl_2 flow was $\sim 4.0 \times 10^{-6}$ mole min^{-1}. The main XeCl band begins at 304 nm and extends to shorter λ; a weaker system (secondary emission) extends to the red. The sharp 236 nm feature is emission from the excited XeCl excimer state. The 256 nm emission in both spectra (but stronger in Ar + Cl_2) arise from molecular Cl_2^*. An additional weaker Cl_2^* emission, super imposed with OH emission, occurs at 306 nm from Ar* + Cl_2. The 199 nm emission from Ar* + Cl_2 is the secondary band of ArCl. Both the broad strong oscillations and the weak sharp oscillations (on the 100K scale) are real effects of the XeCl emission.

Figure 5 - Emission spectra of XeCl* at various pressure: (a) 1.7 torr, (b) 6 torr, (c) 26 torr. For all cases the slits were 55 microns and scan speed was 125 Å min^{-1}. At the highest pressure discrete bands from individual v' levels can be identified at λ = 307.7; 305.7; 304.0; 302.3; 300.7; 298.9,

which correspond to 0-0, 1-3, 2-0, 3-0, 4-0 and 5-0 transitions.[8] The $XeCl(^2\pi_{1/2})$ emission is identified. The unlabeled band at 256 nm is Cl_2^* emission.

Figure 6 - Emission spectra of KrCl* at different total pressures: (a) 1.8 torr, (b) 15 torr, (c) 28 torr. For all cases the slits were 80 microns and the scan speed was 125 Å/min, some emission extend below the short wavelength limit of the monochromator. The Cl_2^* 256 nm band is superimposed on the KrCl secondary emission at low pressure. The sharp band at 198 nm corresponds to emission from the excited KrCl excimer state. Note that the main emission is narrowed and the secondary emission quenched with increasing pressure.

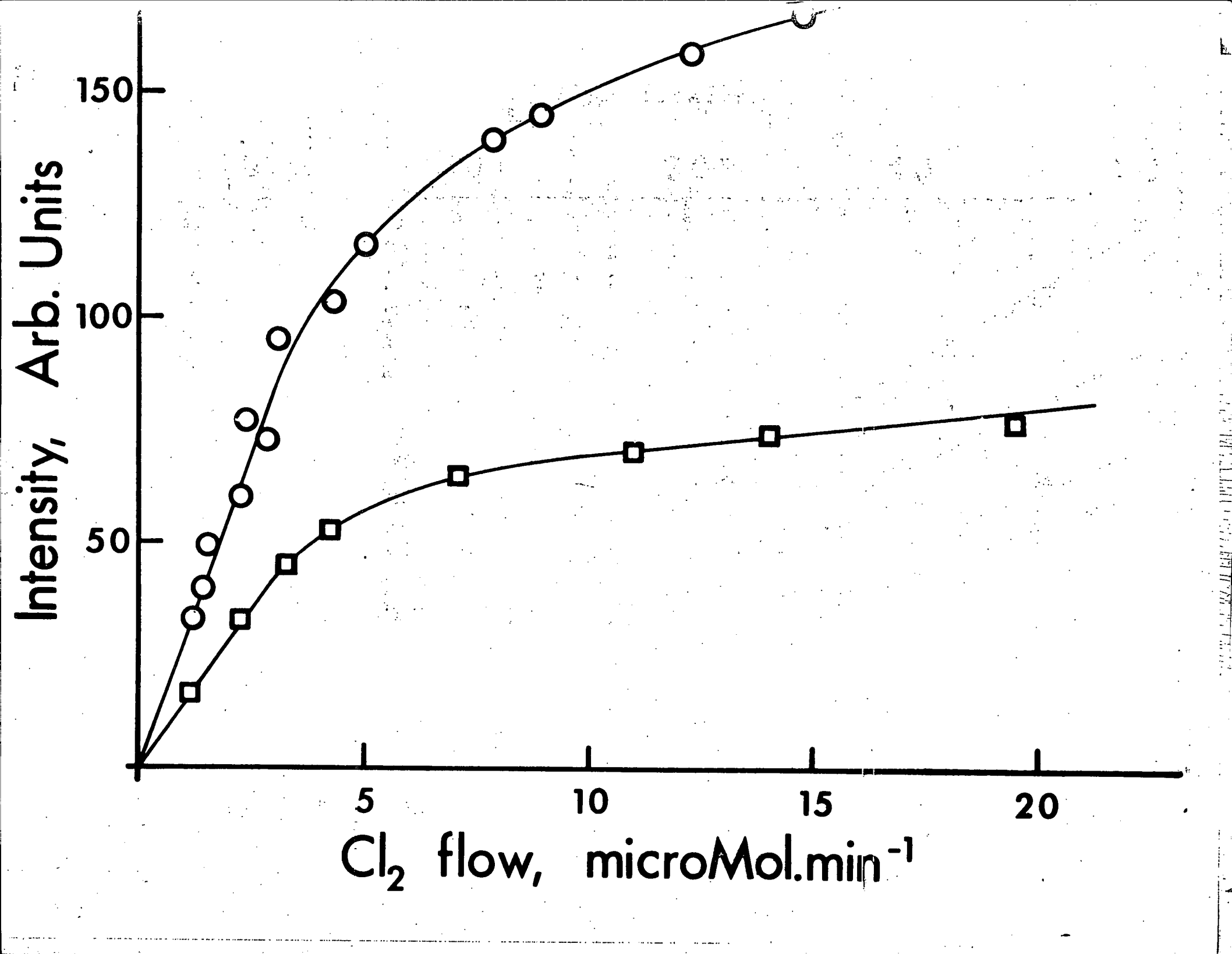
Figure 7 - Emission spectra from $Xe(^3P_2) + N_2F_4$ (a) and $Xe(^3P_2) + BrF_5$ (b) at 2 torr; the latter gives only XeBr. Also shown are the emission spectra from the reaction of $Kr(^3P_2) + F_2$ (d) and $Kr(^3P_2) + ClF_3$ (c) at 1.8 torr. In the latter case, emission from both KrCl* and KrF* are clearly shown. For all cases, the slits were 120 microns. The fast oscillations on the blue side of spectrum (a) are real effects.

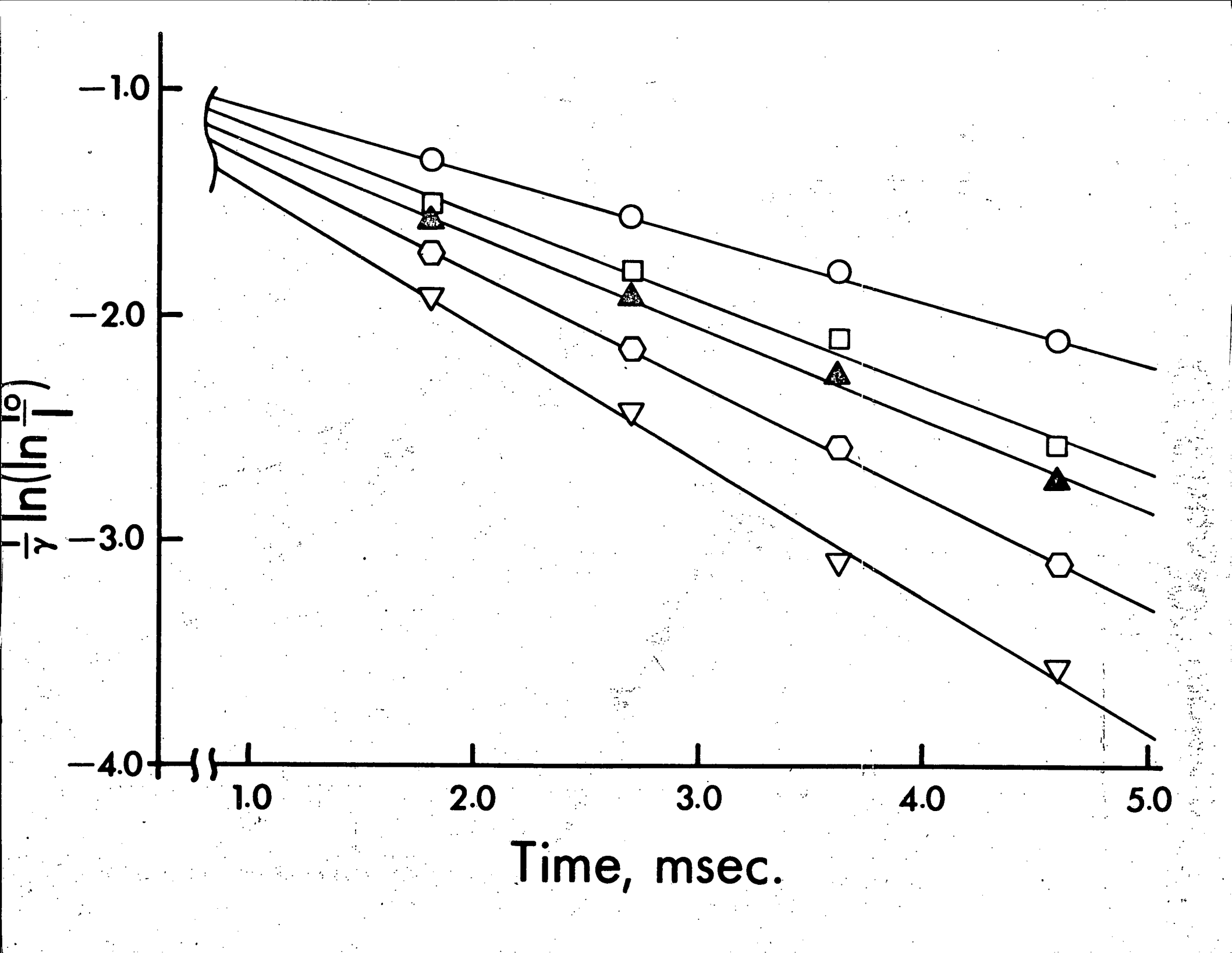
Figure 8 - Emission spectra from $Xe(^3P_2)$ + ONF at various pressures. Notice the extensive relaxation of the main band over the pressure range. The emission from the excited XeF excimer state at 265 nm is still observable at 36 torr.

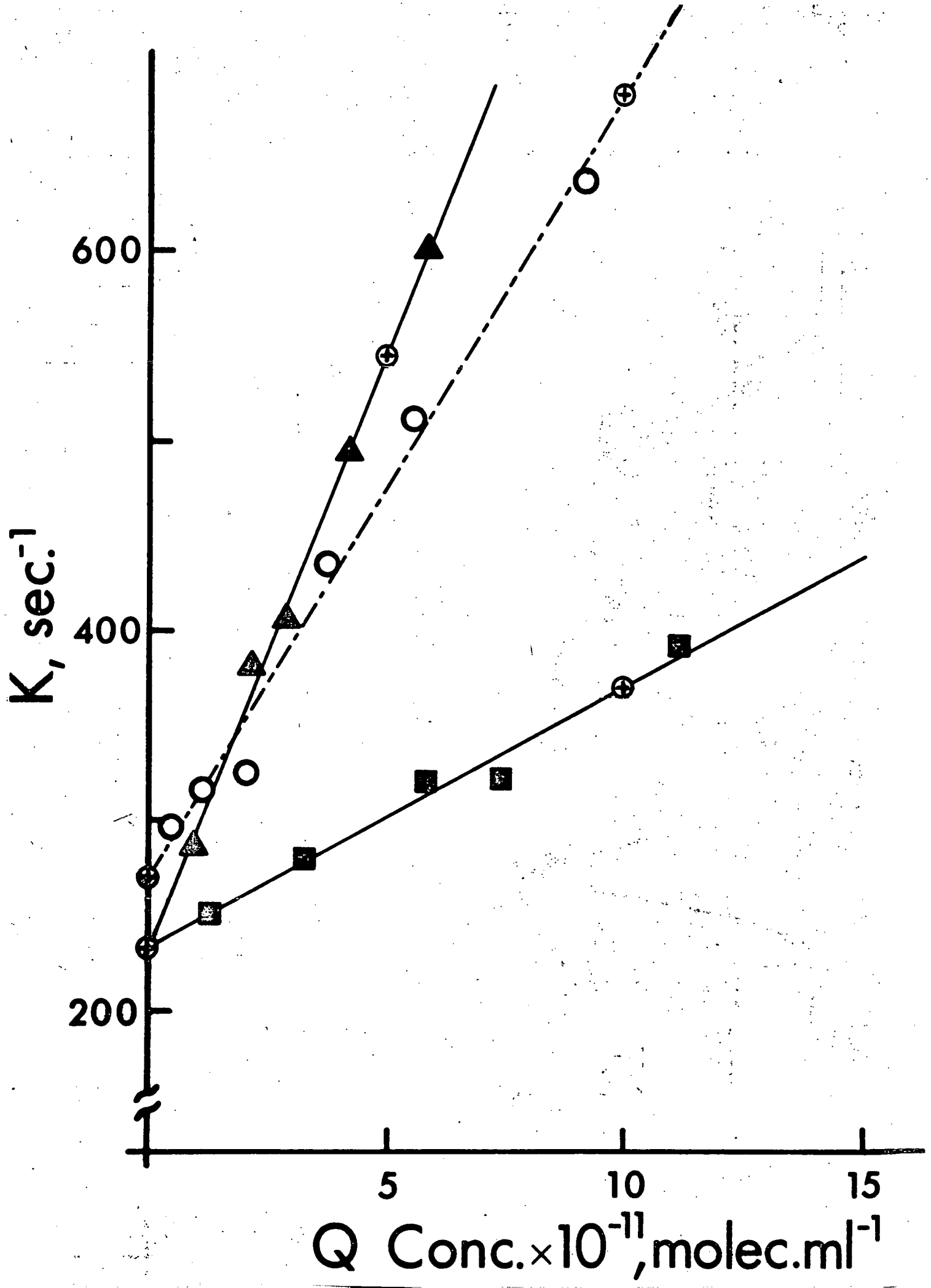
Figure 9 - Emission spectra from $Xe(^3P_2) + ClF_3$ (upper) and XeF_2 (lower) at 1.5 torr. Both XeF and XeCl are formed from ClF_3. Note that the XeF emission from XeF_2 is more relaxed at 1.5 torr than the XeF spectrum at 12 torr from Xe + FNO.

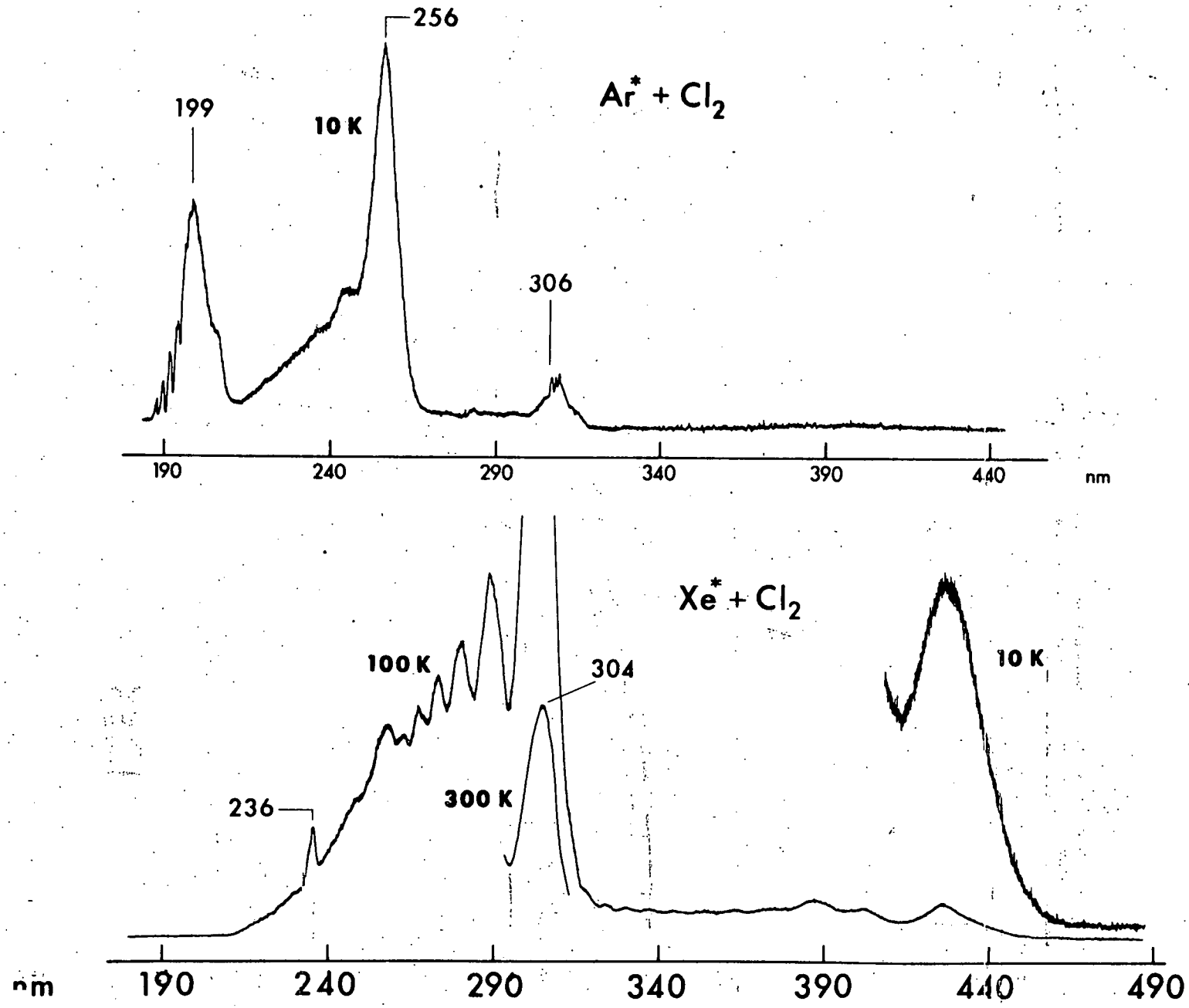
Figure 10 - Calculated KrF potentials. The results were taken from reference 28.

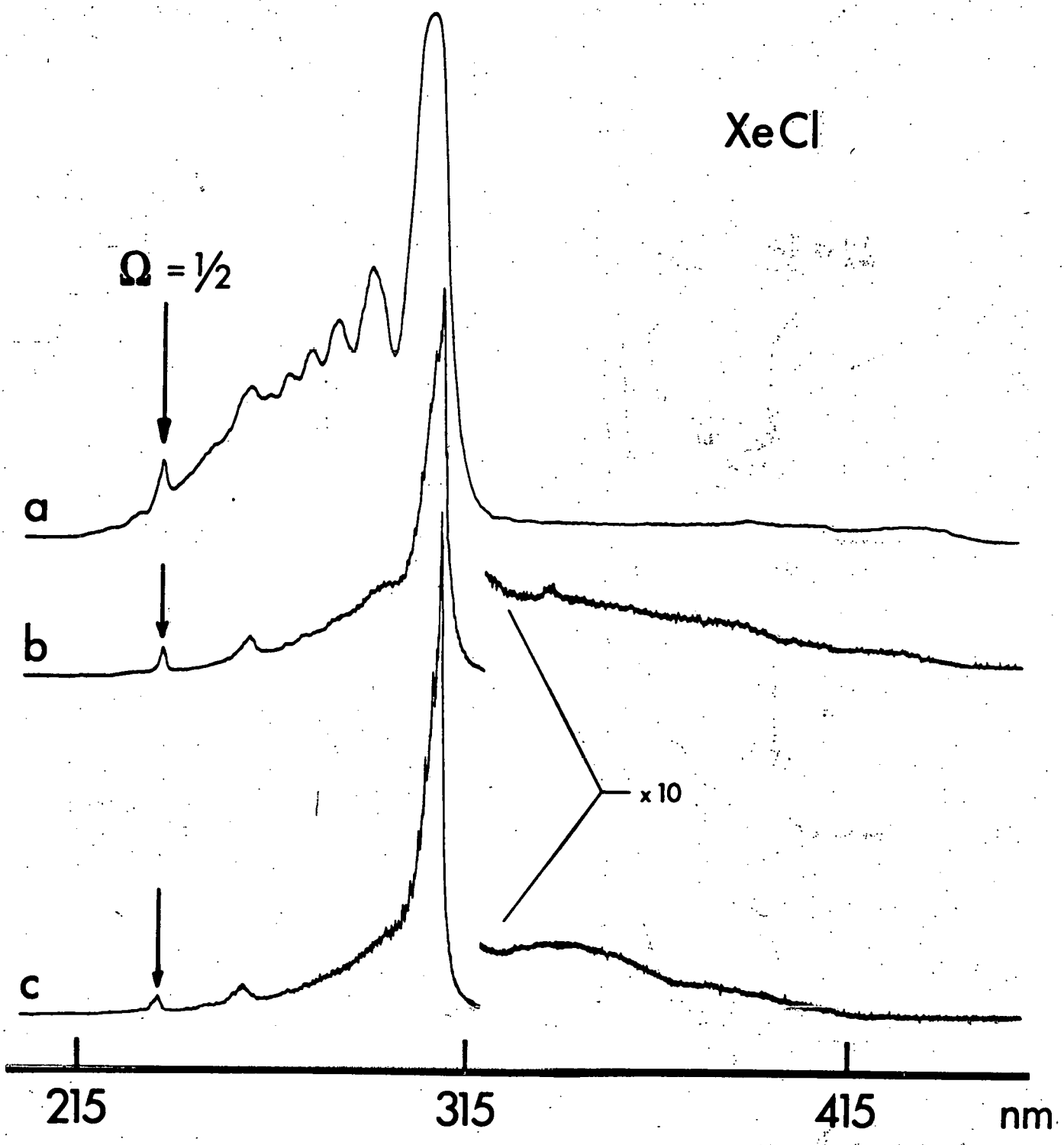
Figure 11 - Metastable $Ar(^3P_2)$ quenching cross sections vs C_6. For convenience of display the scales of the axis are not equal; the line still is of unit slope. The symbols, , denote cross sections measured in this work; the points were previously reported[13] values; they were used here to define the line of unit slope.

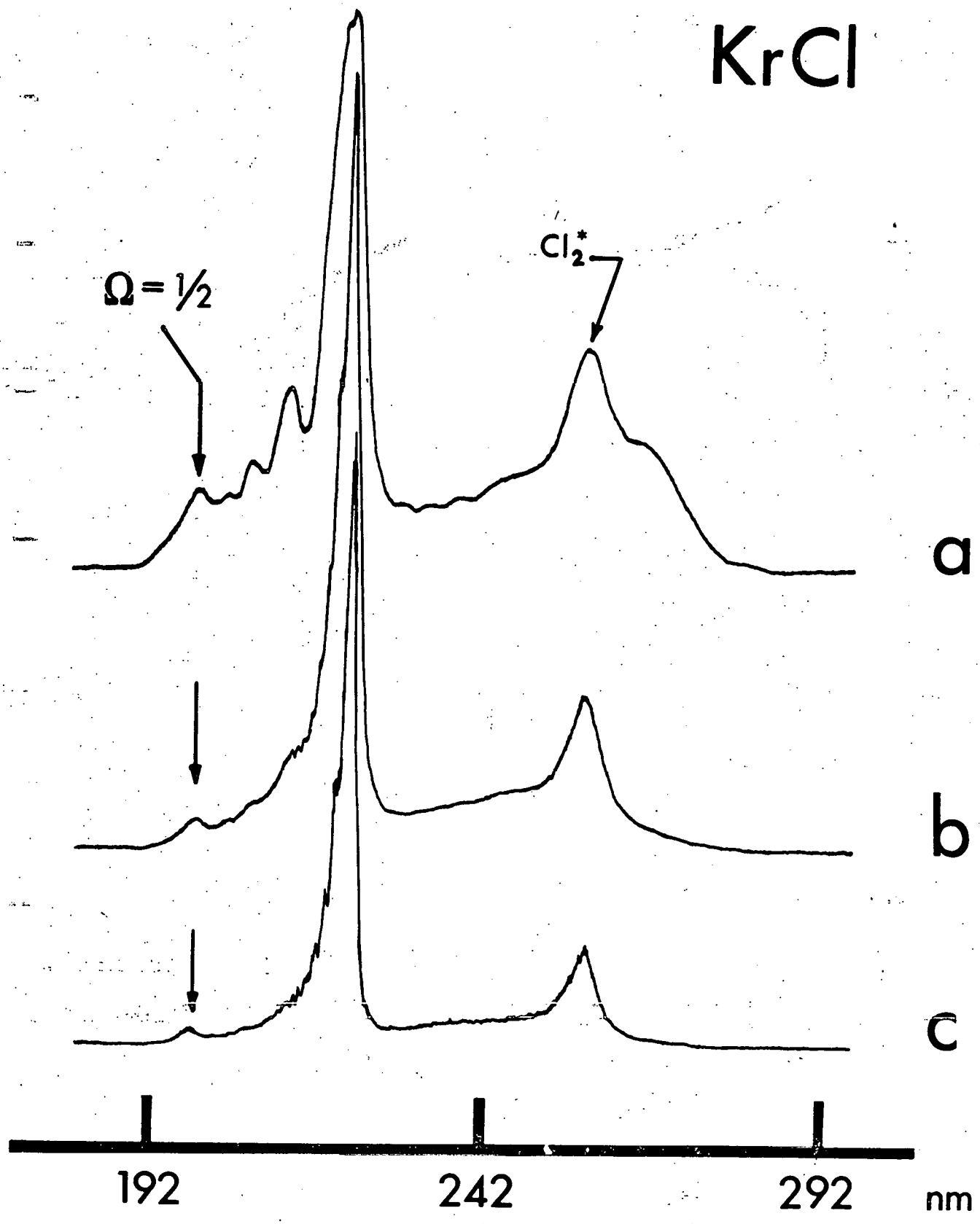

Intensity, Arb. Units
150
100
50
5
10
15
20
Cl2 flow, microMol.min-1

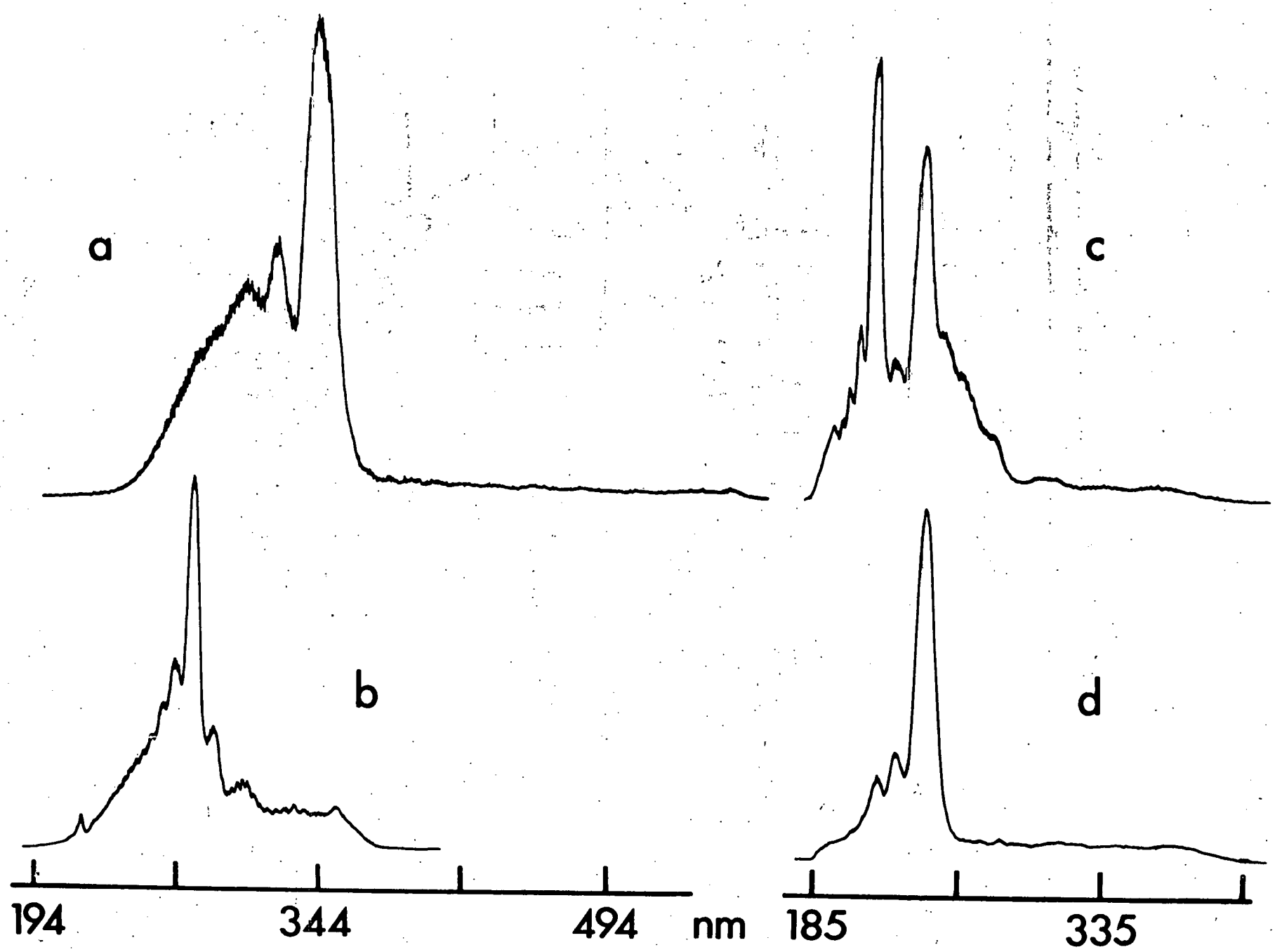

−1.0
−2.0
−3.0
−4.0
$-\frac{1}{\gamma}\ln\left(\ln\frac{I_0}{I}\right)$
1.0
2.0
3.0
4.0
5.0
Time, msec.

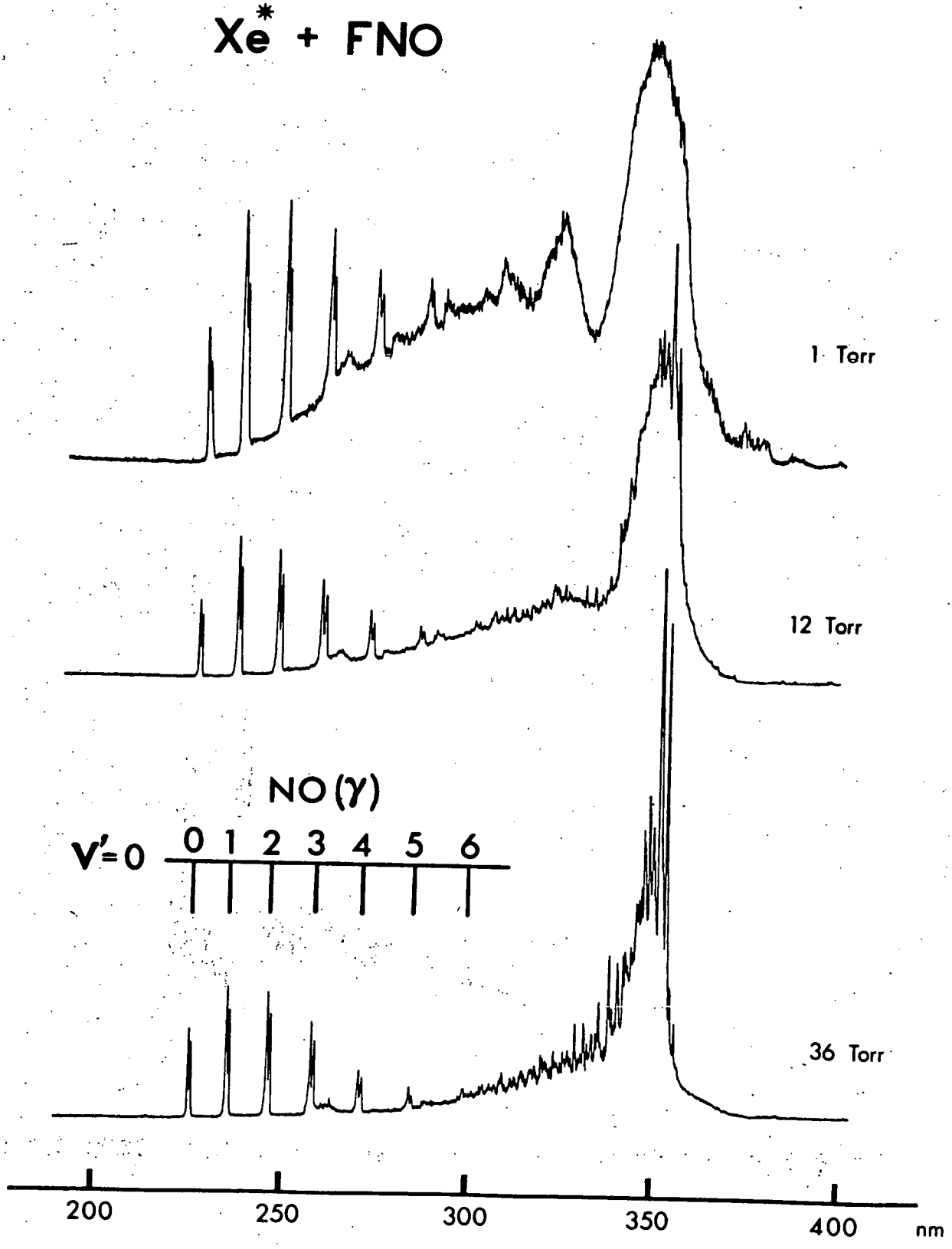

600
400
200
K, sec.-1
5
10
15
Q Conc.×10-11, molec.ml-1

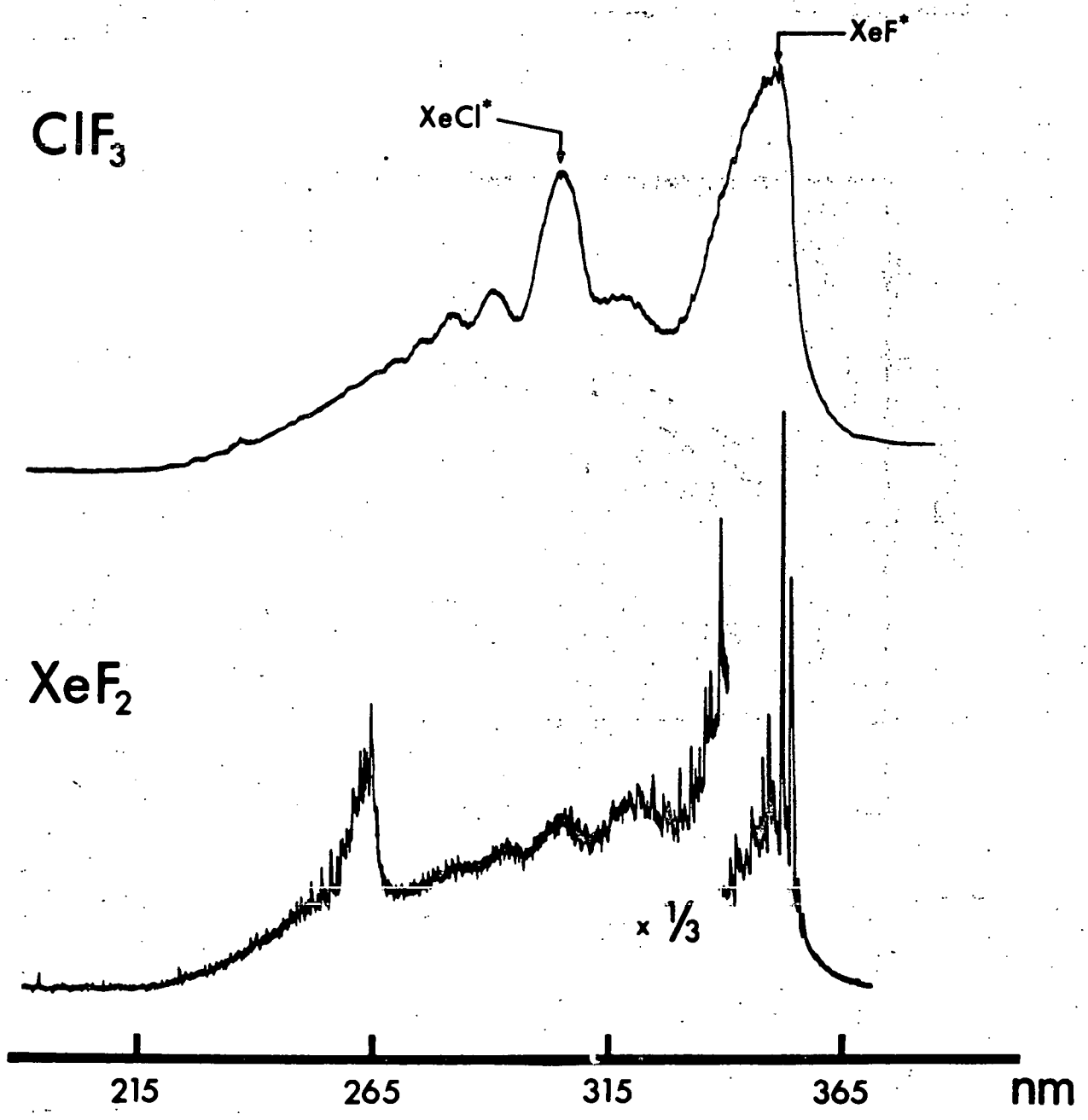

256
199
10 K
Ar* + Cl2
306
190
240
290
340
390
440
nm
Xe* + Cl2
100 K
304
10 K
236
300 K
nm
190
240
290
340
390
440
490


XeCl
Ω = ½
a
b
c
x 10
215
315
415
nm


KrCl
Ω = ½
Cl_2^*
a
b
c
192
242
292
nm


a
b
c
d
194
344
494
nm
185
335

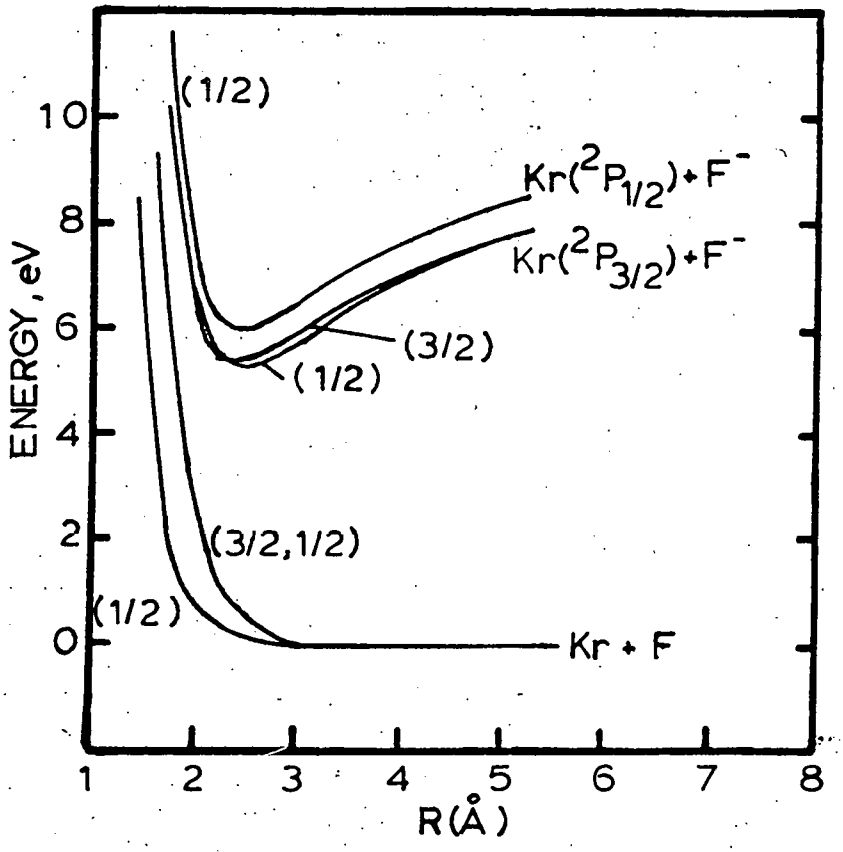

Xe* + FNO
1 Torr
12 Torr
36 Torr
NO(γ)
V'=0
0 1 2 3 4 5 6
200
250
300
350
400
nm

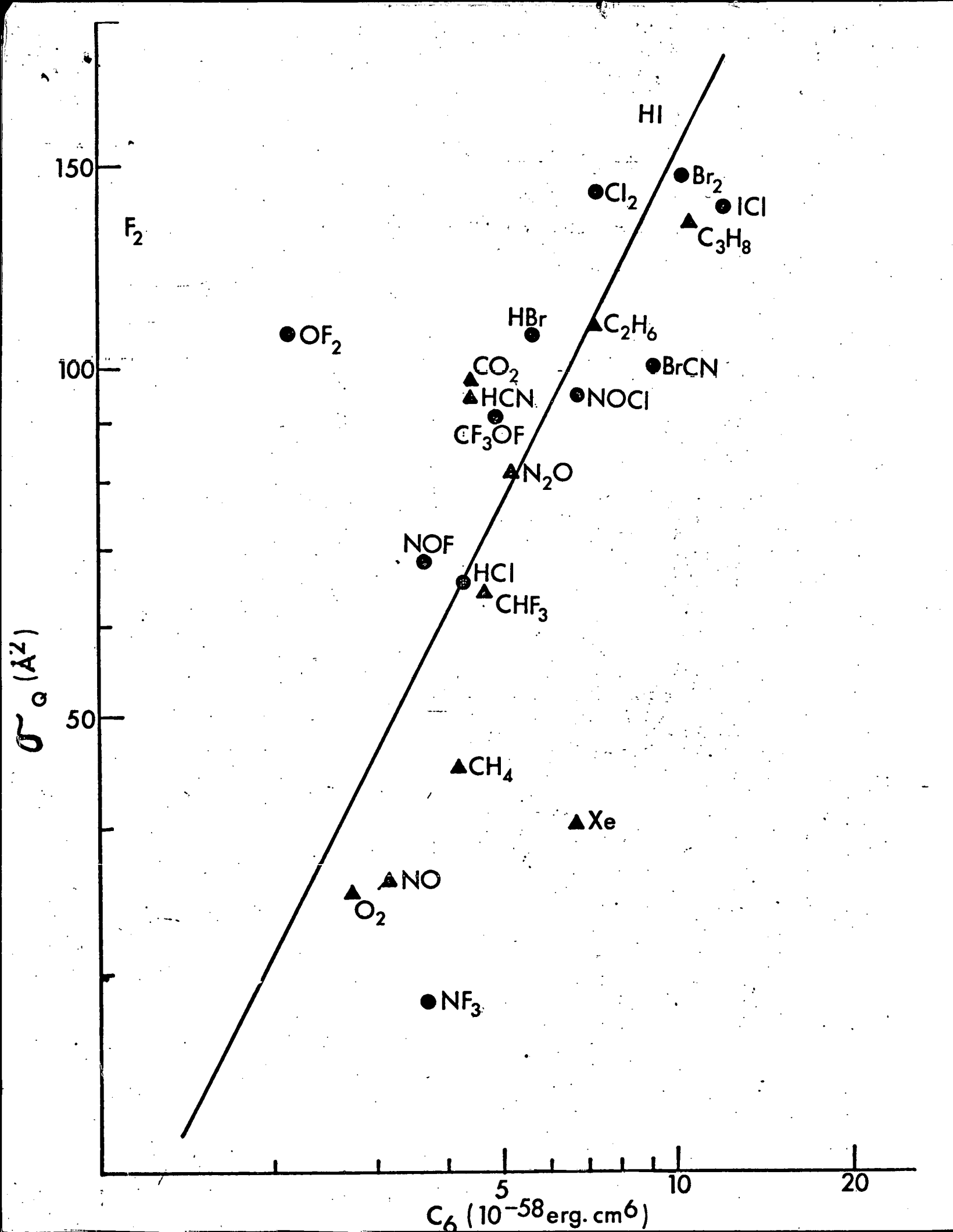

ClF3
XeCl*
XeF*
XeF2
x 1/3
215
265
315
365
nm

Calculated Kr + F potential curves.

[Dunning and Hay, App. Phys. Letts. In Press (1976)]

The "bound" state correlating to $Kr^*(^3P) + F(^2P)$ is not shown on this diagram.




HI
150
F_2
Br_2
Cl_2
ICl
C_3H_8
OF_2
HBr
C_2H_6
100
CO_2
BrCN
HCN
NOCl
CF_3OF
N_2O
NOF
HCl
CHF_3
σ_o (Å^2)
50
CH_4
Xe
NO
O_2
NF_3
5
10
20
C_6 (10^{-58} erg. cm^6)